Assignment:- 8

ALP Program

Write 8086 ALP to find and count negative and positive number from signed array stored in memory and display magnitude of negative numbers.

```
section .data
    arr db 10, -5, -12, 7, -3, 8, 0, -9, 4, -6
    len equ 10
    pos_count db 0
    neg_count db 0
    msg_pos db "Positive count: ", 0
    msg_neg db "Negative count: ", 0
    newline db 10

section .bss
    buffer resb 4     ; for integer to string
conversion

section .text
    global _start

_start:
    mov rcx, len
    lea rsi, [rel arr]

loop_start:
    mov al, [rsi]
    cmp al, 0
    jl negative
    inc byte [pos_count]
    jmp next

negative:
    inc byte [neg_count]

next:
    inc rsi
    loop loop_start

    ; print "Positive count: "
    mov rsi, msg_pos
    call print_string
    movzx rax, byte [pos_count]
    call print_number

    ; newline
    mov rsi, newline
    call print_string

    ; print "Negative count: "
    mov rsi, msg_neg
    call print_string
    movzx rax, byte [neg_count]
    call print_number

    ; newline
    mov rsi, newline
    call print_string

    ; exit
    mov rax, 60
    xor rdi, rdi
    syscall

; ----------------------
; print_string (rsi = address)
print_string:
    push rax
    push rdi
```

```
    push rdx
    mov rax, 1
    mov rdi, 1
    mov rdx, 0
count_loop:
    cmp byte [rsi+rdx], 0
    je print_it
    inc rdx
    jmp count_loop
print_it:
    syscall
    pop rdx
    pop rdi
    pop rax
    ret

; ---------------------
```

```
; print_number (rax = number)
print_number:
    mov rcx, buffer+3
    mov byte [rcx], 0
convert_loop:
    dec rcx
    xor rdx, rdx
    mov rbx, 10
    div rbx
    add dl, '0'
    mov [rcx], dl
    test rax, rax
    jnz convert_loop
    mov rsi, rcx
    call print_string
    ret
```

Output: